

Week 7 Tutorial Solutions

Introduction

This workshop will help you familiarize yourself with the AR time-series model and the EMH.

AR(p) models

Recall that a time series y_t for $t = \{\dots, -1, 0, 1, 2, \dots\}$ is an **AR(p) process** if

$$y_t = c + \phi_1 y_{t-1} + \phi_2 y_{t-2} + \dots + \phi_p y_{t-p} + \varepsilon_t,$$

where $\varepsilon_t \sim WN(0, \sigma_\varepsilon^2)$ (white noise) and $V(\varepsilon_t | \mathcal{F}_{t-1}) = V(\varepsilon_t)$. $\mathcal{F}_{t-1}$ represents all information available to the econometrician at time t .

The AR(p) model is simply a multiple linear regression model with regressors being $y_{t-1}, y_{t-2}, \dots, y_{t-p}$. Therefore, the coefficients $\phi_1, \phi_2, \dots, \phi_p$ can be estimated as

$$(c, \hat{\phi}_1, \dots, \hat{\phi}_p)' = \arg \min \sum_{t=1+p}^T (y_t - c - \phi_1 y_{t-1} - \phi_2 y_{t-2} - \dots - \phi_p y_{t-p})^2.$$

The autocorrelations of the AR(p) process display a constant pattern of decay. For example, in the AR(1) model, the autocorrelations decay as ϕ_1^k , for $k = 1, 2, \dots$.

Question 1

Consider an AR(1) process given by

$$y_t = \phi_1 y_{t-1} + \varepsilon_t,$$

where $\varepsilon_t \sim_{iid} WN(0, \sigma_\varepsilon^2)$ (white noise). In order for this process to be stationary we require that condition $|\phi_1| < 1$ be met. Even when $|\phi_1| < 1$, the absolute value of ϕ_1 determines the ‘persistence’ of the process. To illustrate this point:

1. Run the code below to generate 3 series of 500 observations of an AR(1) process where $\phi_1 = 0.1, 0.8$ and -0.8 respectively. Plot each of these series. Briefly comment on these realisations.

```
set.seed(12345678) #so we all get the same answers!

n <- 500

#####
# phi = 0.1
phi <- 0.1

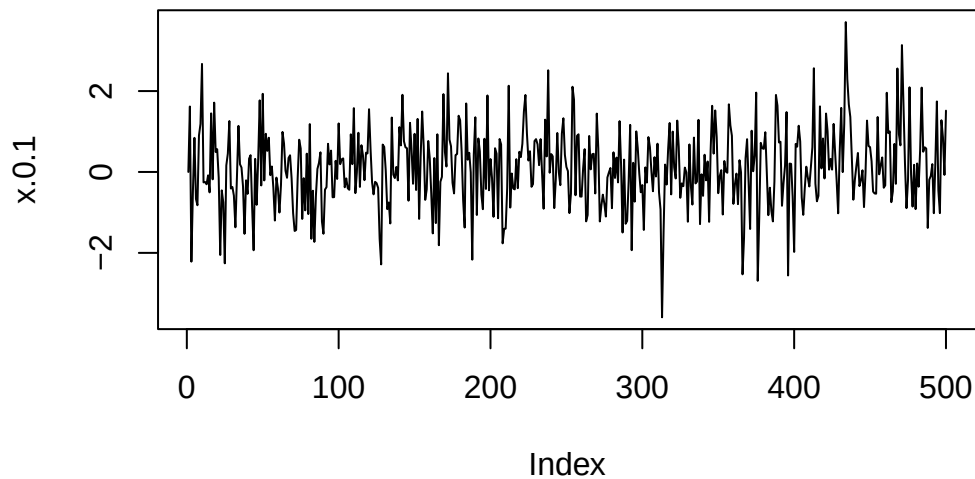
eps.0.1 <- rnorm(n, mean = 0, sd = 1)

x.0.1 <- numeric(n)
x.0.1[1] <- 0 # initial value

for (t in 2:n) {
  x.0.1[t] <- phi * x.0.1[t - 1] + eps.0.1[t]
}

plot(x.0.1, type = "l", main = "AR(1) phi = 0.1")
```

AR(1) phi = 0.1



```
#####
# phi = 0.8
```

```

phi <- 0.8

eps.0.8 <- rnorm(n, mean = 0, sd = 1)

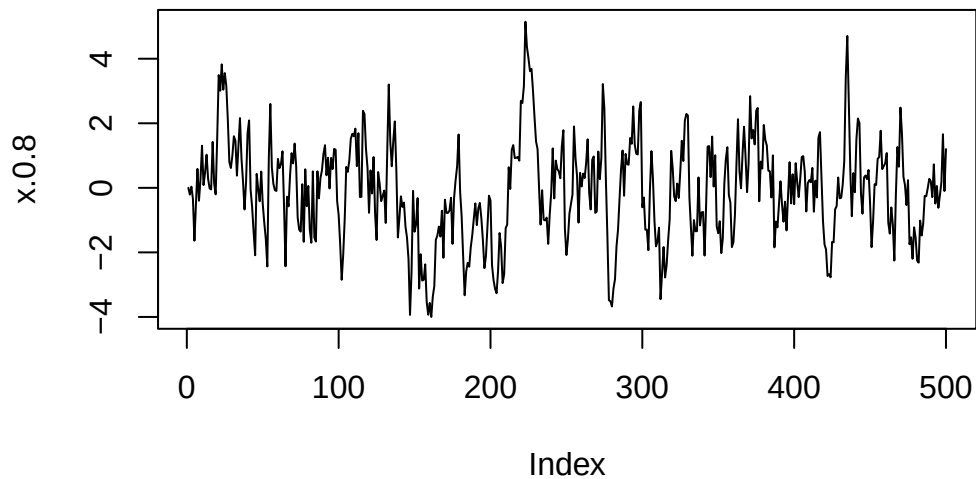
x.0.8 <- numeric(n)
x.0.8[1] <- 0 # initial value

for (t in 2:n) {
  x.0.8[t] <- phi * x.0.8[t - 1] + eps.0.8[t]
}

plot(x.0.8, type = "l", main = "AR(1) phi = 0.8")

```

AR(1) $\phi = 0.8$



```

#####
# phi = -0.8
phi <- -0.8

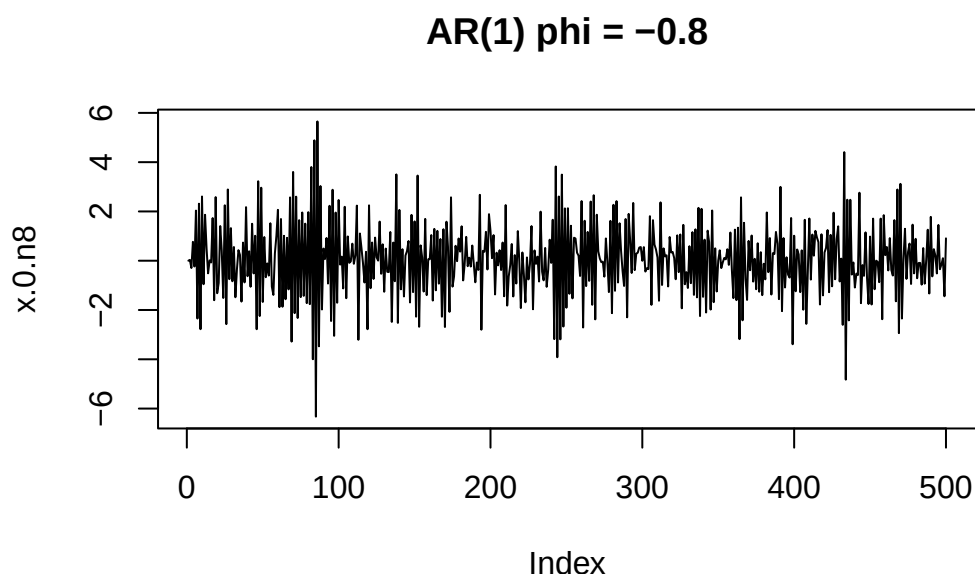
eps.0.n8 <- rnorm(n, mean = 0, sd = 1)

x.0.n8 <- numeric(n)
x.0.n8[1] <- 0 # initial value

for (t in 2:n) {
  x.0.n8[t] <- phi * x.0.n8[t - 1] + eps.0.n8[t]
}

```

```
plot(x.0.n8, type = "l", main = "AR(1) phi = -0.8")
```



```
# optional
# if you want to put dates on the simulated data, using xts object
# library(xts)
#dates <- seq.Date(from = as.Date("2020-01-01"), by = "day", length.out = n)
#x.0.1.xts <- xts(as.numeric(x.0.1), order.by = dates)
#colnames(x.0.1.xts) <- "x phi 0.1"

#head(x.0.1.xts)
#plot(x.0.1.xts, main = "Simulated AR(1) phi = 0.1")
```

Notice that when persistence is low ($\phi_1 = 0.1$), the line plot is more jagged. The series does not stay above or below 0 for long. When $\phi_1 = 0.8$, there are longer periods where the series stays above and below 0, indicating stronger persistence. When $\phi_1 = -0.8$, the persistence shows as clustering. Each positive observation must be followed with a negative one, so the graph crosses 0 each time, but the width of the value is similar. Most of the time in econometrics, if there is persistence, $\phi_1 > 0$, but it must be strictly less than one in absolute value to be (weakly) stationary (*i.e.* $|\phi_1| < 1$).

5. Use the code below to produce the correlograms for each of the time series created above. Briefly discuss.

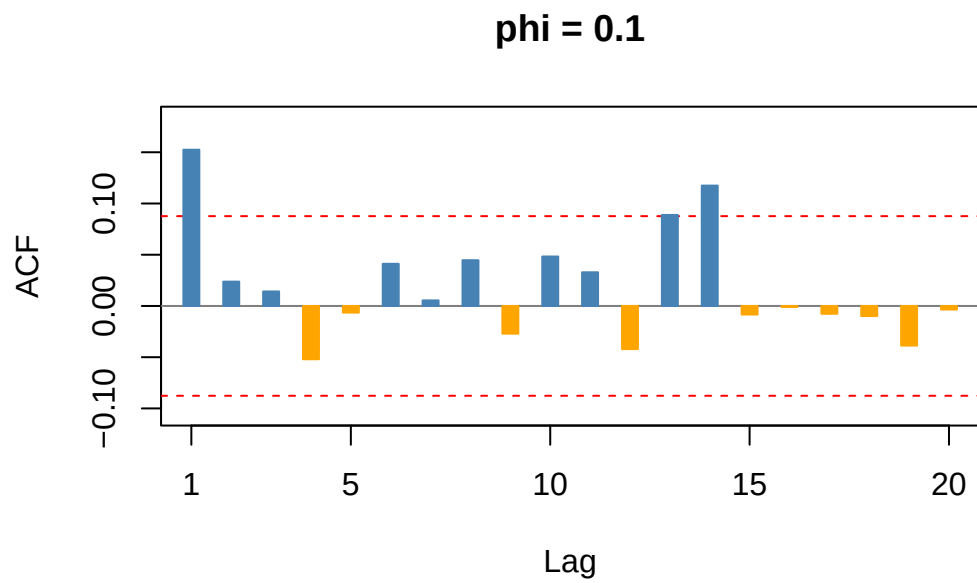
Identifying persistence is easier if we use the correlograms. In the ACF, if there is persistence with **positive** correlation, the spikes will gradually approach zero in a smooth (wavy) way.

The stronger the persistence, the longer it takes to become zero (like a fence). If there is persistence with **negative** correlation, we will have alternating spikes.

```
# Use the Correlogram.R function from last week

source("Correlogram.R")
correlogram(x.0.1,20,main.acf = "phi = 0.1")
```

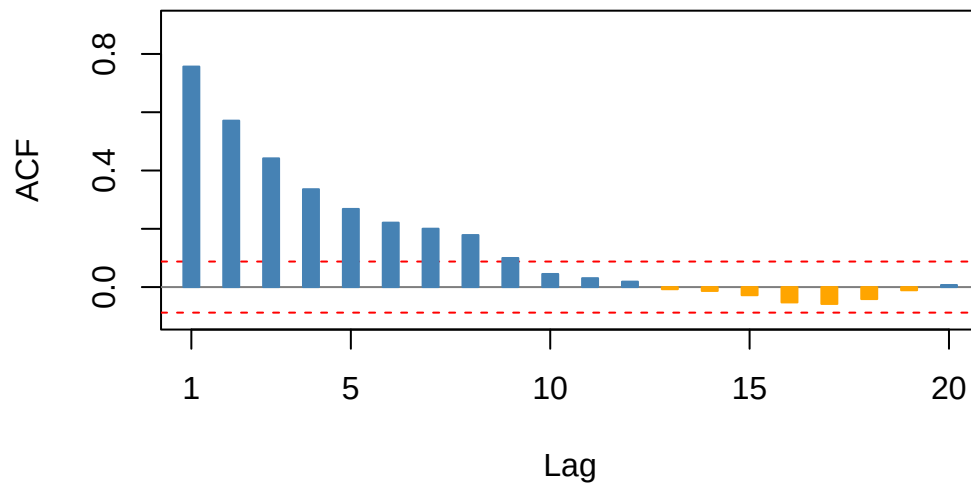
Lag	AC	Q	Prob
1	0.1525	11.6924	0.0006276
2	0.0238	11.9768	0.0025080
3	0.0142	12.0782	0.0071200
4	-0.0520	13.4469	0.0092860
5	-0.0065	13.4681	0.0193700
6	0.0412	14.3287	0.0261700
7	0.0054	14.3437	0.0453900
8	0.0446	15.3606	0.0525000
9	-0.0270	15.7339	0.0726500
10	0.0483	16.9272	0.0759900
11	0.0329	17.4827	0.0943900
12	-0.0421	18.3942	0.1042000
13	0.0888	22.4622	0.0486000
14	0.1175	29.5907	0.0086850
15	-0.0084	29.6271	0.0133400
16	-0.0010	29.6275	0.0200300
17	-0.0076	29.6578	0.0289200
18	-0.0098	29.7081	0.0403800
19	-0.0387	30.4878	0.0459100
20	-0.0034	30.4940	0.0622300



```
correlogram(x.0.8,20,main.acf = "phi = 0.8")
```

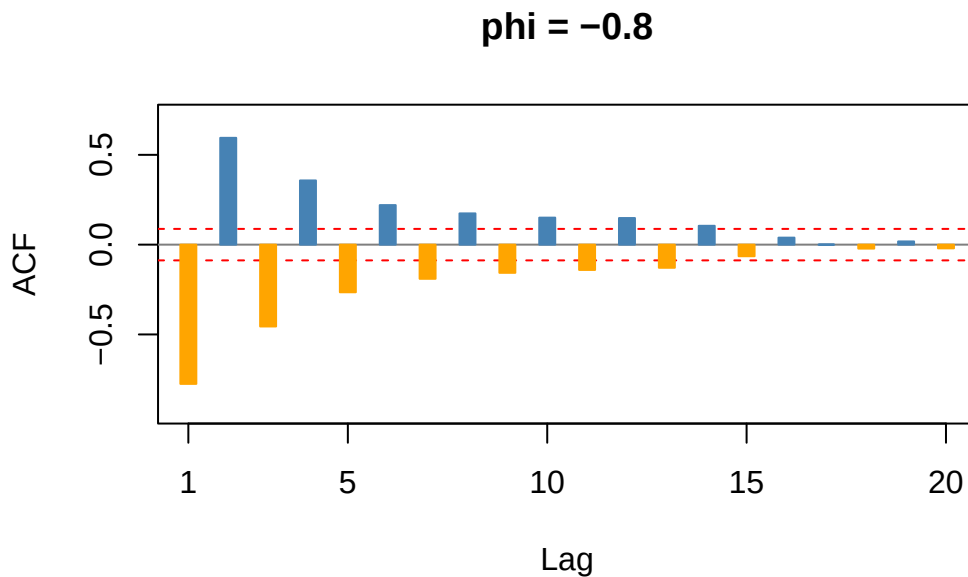
Lag	AC	Q	Prob
1	0.7566	287.9559	1.387e-64
2	0.5710	452.3098	6.056e-99
3	0.4417	550.8332	4.585e-119
4	0.3358	607.8933	3.033e-130
5	0.2682	644.3541	5.258e-137
6	0.2211	669.1872	2.744e-141
7	0.2003	689.6151	1.195e-144
8	0.1785	705.8650	3.908e-147
9	0.0998	710.9577	3.048e-147
10	0.0449	711.9911	1.673e-146
11	0.0306	712.4708	1.144e-145
12	0.0186	712.6495	8.636e-145
13	-0.0073	712.6767	6.714e-144
14	-0.0132	712.7673	4.854e-143
15	-0.0279	713.1707	2.896e-142
16	-0.0526	714.6050	1.007e-141
17	-0.0578	716.3423	2.923e-141
18	-0.0411	717.2239	1.253e-140
19	-0.0108	717.2850	7.797e-140
20	0.0071	717.3111	4.800e-139

phi = 0.8



```
correlogram(x.0.n8, max.lag = 20 , main.acf = "phi = -0.8")
```

Lag	AC	Q	Prob
1	-0.7753	302.3778	9.994e-68
2	0.5946	480.5896	4.378e-105
3	-0.4556	585.4106	1.466e-126
4	0.3574	650.0348	2.291e-139
5	-0.2651	685.6729	6.150e-146
6	0.2201	710.2858	3.678e-150
7	-0.1902	728.7020	4.463e-153
8	0.1739	744.1267	2.250e-155
9	-0.1568	756.6996	4.424e-157
10	0.1503	768.2693	1.364e-158
11	-0.1408	778.4445	8.035e-160
12	0.1481	789.7247	2.643e-161
13	-0.1288	798.2793	3.227e-162
14	0.1049	803.9668	1.568e-162
15	-0.0645	806.1194	4.200e-162
16	0.0390	806.9079	2.128e-161
17	0.0002	806.9079	1.537e-160
18	-0.0219	807.1581	9.515e-160
19	0.0178	807.3233	5.967e-159
20	-0.0206	807.5447	3.541e-158



- When $\phi = 0.1$, the spikes go to zero quickly, then become significant again around observations 10-15, in a small wave. This is typical of weaker persistence.
- When $\phi = 0.8$, persistence is stronger and the spikes gradually go to 0 by about observation 8.
- When $\phi = -0.8$, persistence is strong as it is identical to when $\phi = 0.8$, except it has alternating positive and negative spikes.

Typically, we observe positive correlations (if any). Using the correlograms to identify patterns and check for persistence is important for the rest of the unit.

Question 2

An implication of the Efficient Market Hypothesis (EMH) is that there should be no predictable pattern in financial log returns; that is, they should behave randomly over time.

Now consider the hourly observations of the US dollar–Deutsche Mark (*USD/DM*) exchange rate from 00:00 on 1 January 1986 to 11:00 on 15 July 1986. These data are available in the file [usd_dm.xlsx](#) on Moodle.

- Compute the log returns on the *USD/DM* exchange rate.
- True or False? The *USD/DM* foreign exchange market is efficient. Explain your answer.

First, let's plot the data and then the log return series (defined as $r_t = \log(1 + R_t)$ where R_t is the simple return. Run the code below.

```

library(xts)

usdm <- readxl::read_excel("usd_dm.xlsx")

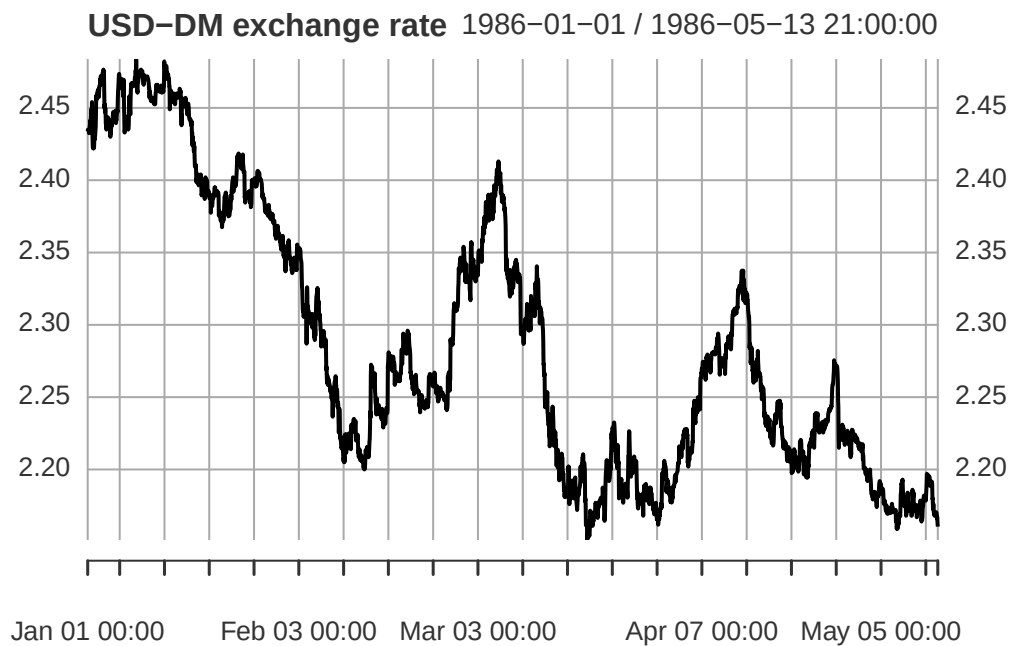
# Convert datetime column to POSIXct
usdm$datetime <- as.POSIXct(usdm$datetime, tz = "UTC")

# Convert to xts
usd_dm.xts <- xts(usdm[, setdiff(names(usdm), "datetime")],
                  order.by = usdm$datetime)

# log return
logret <- na.omit(diff(log(usd_dm.xts)))

# plot
plot(usd_dm.xts, main = "USD-DM exchange rate")

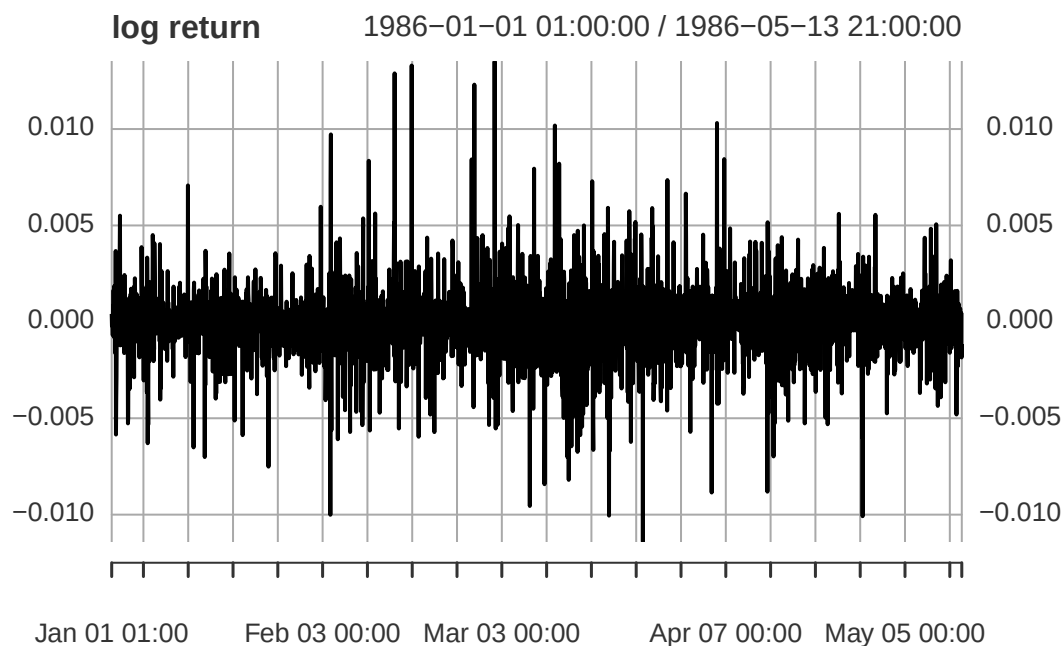
```



```

# plot log return
plot(logret, main = "log return")

```



The exchange rate series is not stationary, however the log-returns are. This is revision, but we need to always perform a visual check - even if we don't use it for anything (*e.g.* in a report).

Let $\gamma(k)$ be the autocorrelation function of r_t .

Test of EMH:

$$H_0 : \gamma(k) = 0$$

$$H_1 : \gamma(k) \neq 0$$

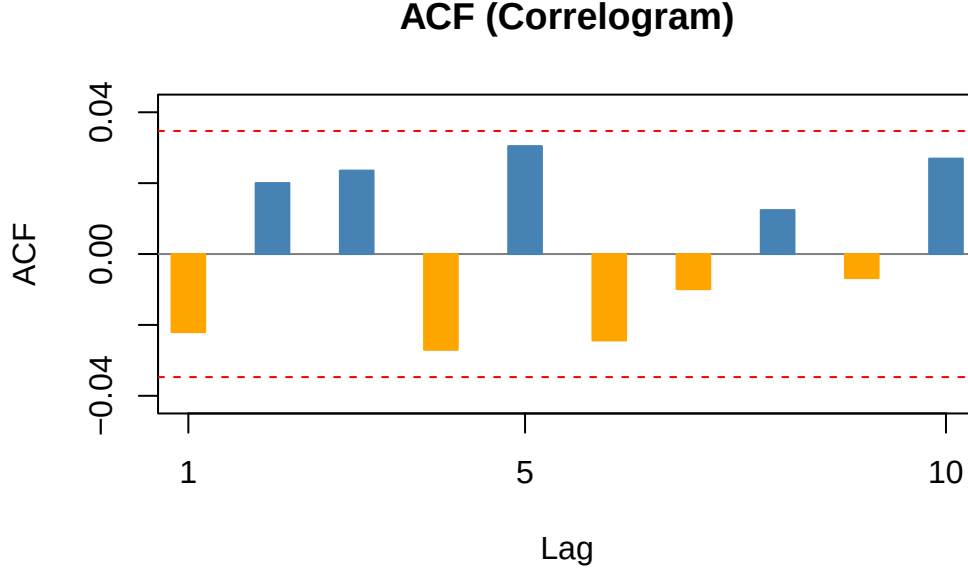
The test statistic = $\sqrt{T}\hat{\gamma}(k)$. We reject H_0 if $|\sqrt{T}\hat{\gamma}(k)| > 1.96$.

Run the code below to get the $\hat{\gamma}(k)$'s and the correlogram for the log-returns.

```
source("Correlogram.R")
# in the correlogram function "fit = NULL"
correlogram(logret,max.lag = 10)
```

Lag	AC	Q	Prob
1	-0.0220	1.5493	0.21320
2	0.0200	2.8303	0.24290
3	0.0235	4.6000	0.20350
4	-0.0270	6.9329	0.13950
5	0.0304	9.8962	0.07823
6	-0.0244	11.7965	0.06666
7	-0.0099	12.1110	0.09697
8	0.0124	12.6028	0.12630

9	-0.0068	12.7495	0.17430
10	0.0269	15.0689	0.12960



The EMH can be tested by testing $\hat{\gamma}(1)$ from the correlogram (column AC). We have $USD/DM : \hat{\gamma}(1) = -0.022$, with a calculated test statistic $= |\sqrt{3189} * (-0.022)| = 1.2424$. Since this is < 1.96 , we cannot reject H_0 at the 5% level of significance (in fact, none of the $\hat{\gamma}(i)$'s in the correlogram are significantly different from zero).

NOTE: this is similar to the Q-stat in the table (the Ljung-Box test), but they are not exactly the same - hence the difference in values, but not the decision.

We cannot reject the EMH for log returns of USD/DM exchange rate. This implies that the returns are not predictable and hence these results are consistent with the EMH.

We can also use the BG test for autocorrelation of the residuals (we will test for serial correlation up to order 2):

$$\hat{\varepsilon}_t = r_t - \bar{r}, \quad \bar{r} = \frac{1}{T} \sum_{t=1}^T r_t.$$

Using the auxiliary regression model

$$\hat{\varepsilon}_t = \gamma_0 + \gamma_1 \hat{\varepsilon}_{t-1} + \gamma_2 \hat{\varepsilon}_{t-2} + v_t$$

where v_t is some random disturbance term.

$$H_0 : \gamma_1 = \gamma_2 = 0 \quad [\text{EMH is satisfied}]$$

$$H_1 : \gamma_1 \text{ and/or } \gamma_2 \neq 0 \quad [\text{EMH is NOT satisfied}]$$

From the auxiliary regression model:

$$\hat{\varepsilon}_t = \gamma_0 + \gamma_1 \hat{\varepsilon}_{t-1} + \gamma_2 \hat{\varepsilon}_{t-2} + v_t,$$

the test statistic $W = T \cdot R^2 \sim \chi_2^2$ under H_0 .

```
# LM test
library(lmtest)
fit <- lm(logret ~ 1) # run this to generate errors for LM test
bptest(fit, order = 2, type = "Chisq")
```

Breusch-Godfrey test for serial correlation of order up to 2

```
data: fit
LM test = 2.7667, df = 2, p-value = 0.2507
```

The p-value of the BG test is 0.2507. This.....

Question 3

Next, consider the monthly stock returns of the Coca-Cola Company (ticker symbol: KO) from January 1961 to September 2011. The simple returns are stored in the file `coca_cola.xlsx`.

Run the code below to create log-returns and convert to `xts`. Copy and modify the code from Q2 to answer the question. A plot of the log returns is below.

```
coke <- readxl::read_excel("coca_cola.xlsx") # <-- change filename

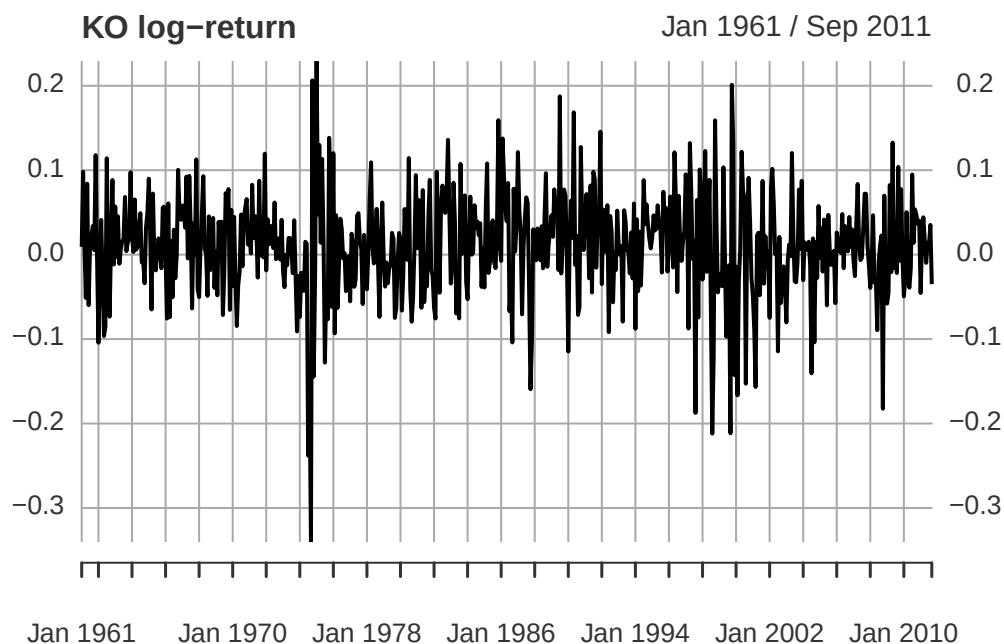
# ignore the date column
coke = coke[,2]

# Jan 1961 to Sept 2011
idx <- seq(from = as.Date("1961-01-01"), by = "month",
           length.out = nrow(coke))
KO.xts <- xts(coke, order.by = idx)

# just display month-year without date
index(KO.xts) <- as.yearmon(index(KO.xts)) # match by year-month

# Transform the simple returns to log returns
logret1 <- log(1+KO.xts)

# plot
plot(logret1, main = "KO log-return")
```



- True or False? The log returns of KO are consistent with the EMH. Explain your answer.

Run the code below to create log-returns and convert to `xts`. Copy and modify the code from Q2 to answer the question. Also, plot the log returns.

Put code chunk here

The correlogram is:

Put code chunk here

Do the BG test like in Q2 and write full answers here.